COE304-F02

College of Engineering & Computer Science

Computer Engineering Department

DRIVER WARNING DEVICE

Introduction to Embedded Systems [COE-304]

جامعة المستقبل
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ABSTRACT

This project develops an Arduino-powered "Driver Warning Device" using an SW520D tilt sensor to monitor driver head position and trigger alerts for unusual behavior, addressing the critical issue of driver distraction. The system, with a tilt sensor, Arduino Nano, and buzzer, is designed to improve road safety by delivering real-time alerts and reducing driver distraction-related Accidents.

INTRODUCTION

Driver distraction has become a significant contributor to road accidents. This project introduces an affordable, and efficient solution designed to address this challenge and enhance driver awareness. Powered by Arduino, the system uses an SW520D tilt sensor to monitor head position and detect signs of drowsiness or distraction, by providing timely alerts, our device aims to reduce accidents and improve road safety effectively.

OBJECTIVES

The objectives of the project are as follows:

1. Enhance driver awareness by providing timely alerts.
2. Monitor driver alertness using a tilt sensor to detect unusual head movements.
3. Utilize Arduino technology to ensure reliable and accurate data processing.
4. Develop a low-cost, portable system suitable for various vehicles.
5. Implement real-time alerts to reduce accident risks caused by distraction or fatigue.
6. Explore potential enhancements for future versions, such as integrating additional sensors and advanced features.

MATERIALS & METHODS

Materials:

Arduino Nano, SW520D tilt sensor, Buzzer, Breadboard, 9V battery, and Jumper Wires.

Methods:

Components were assembled on a breadboard and programmed using Arduino IDE, the system detects head tilts and triggers a buzzer when exceeding a threshold.

Figure 1. Arduino IDE

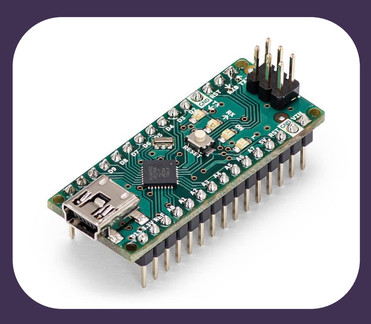

Figure 2. Arduino Nano

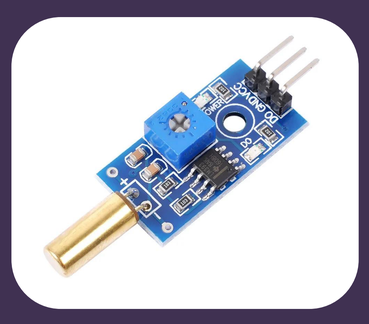

Figure 3. SW520D tilt sensor

RESULTS & DISCUSSION

The device was tested for hardware and software functionality, hardware tests confirmed the accurate detection of head tilts and clear buzzer output. Software testing ensured the Arduino code properly monitored sensor data and triggered the buzzer.

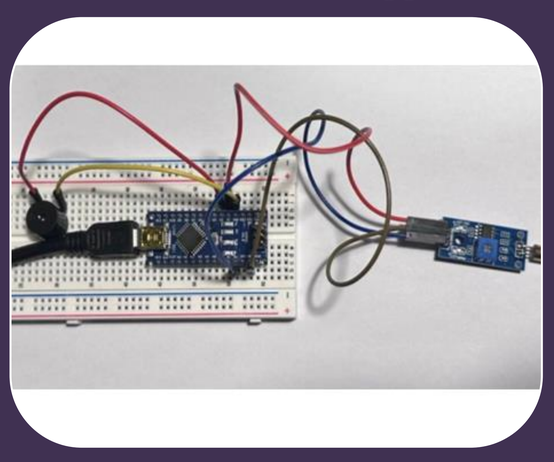

Figure 1. Project Prototype

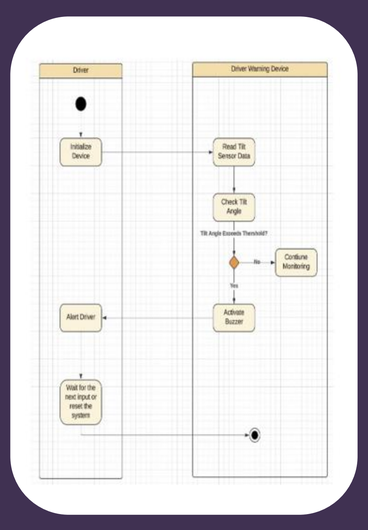

Figure 1. UML Activity Chart

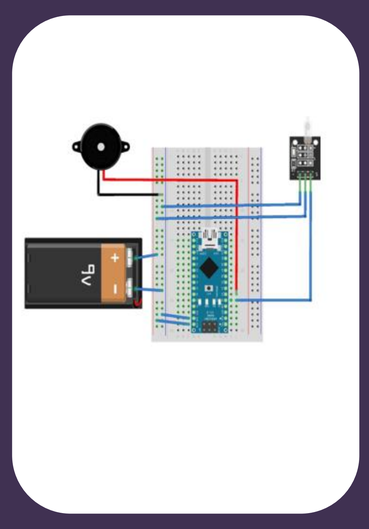

Figure 2. Circuit Diagram

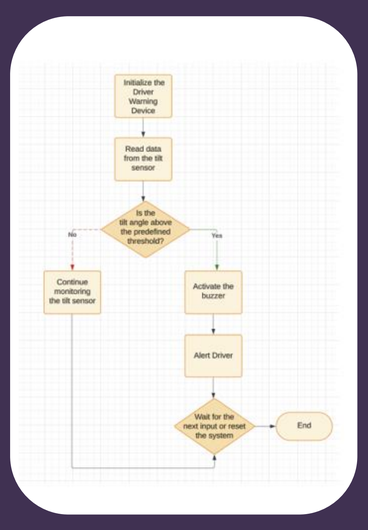

Figure 3. Flowchart

CONCLUSION

The Driver Warning Device effectively addressed the pressing issue of driver distraction and fatigue by achieving accurate detection of unsafe head positions and providing timely alerts. Through successful hardware and software testing, it demonstrate reliability in enhancing road safety. This simple, cost-effective solution shows great potential in reducing accidents.

REFERENCES

- Arduino Nano. *Getting Started with the Arduino Nano*. Article available at: https://www.arduino.cc/en/Guide/ArduinoNano
- P. Sandeep Chary, S. Pranay, N. Sai Kishore, M. Ravi Kumar. Anti Sleep Alarm for Drivers. *Journal of Engineering Sciences*, vol. 14, no. 06, pp. 257-260, 2023.